

Saptarshi Debnath

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Visionary Technical Architect and Staff Engineer with expansive expertise in designing, scaling, and modernizing high-latency, cloud-native distributed systems. Proven track record of leading architectural transformations, driving organizational engineering standards, and mentoring cross-functional teams to deliver highly available, zero-trust infrastructure. Seeking to leverage deep expertise in system design and microservices as a Principal Architect to drive enterprise-wide technical strategy.

EXPERIENCE

Staff Engineer & System Design Architect , *Marriott International*

2020 - Present

- Architecting and defining the technical vision for NextGen distributed systems serving millions of daily users with single-digit millisecond latency.
- Spearheading the migration of monolithic legacy systems to resilient, containerized, cloud-native microservices using Spring Boot, AWS, and Kubernetes.
- Establishing engineering standards and asynchronous communication patterns (Kafka) across multiple engineering pods.
- Mentoring senior engineering teams on system design, distributed data patterns, operational excellence, and CI/CD best practices.

Senior Software Engineer , *Marriott International*

2017 - 2020

- Designed and developed highly scalable, secure, and fault-tolerant RESTful microservices for enterprise integration.
- Optimized JVM performance, database queries, and cloud resource allocation, reducing infrastructure costs by 15%.
- Implemented circuit breakers, bulkheads, and centralized logging to ensure robust system observability and prevent cascading failures.
- Collaborated with product and infrastructure teams to streamline DevOps workflows utilizing Jenkins, Git, and Docker.

Senior Software Engineer , *Wipro*

Oct 2015 - Sep 2017

- Engineered full-stack solutions and orchestrated deployment pipelines using Java, Spring, Hibernate, and AWS.
- Drove Agile adoption by integrating automated testing (Cucumber) and continuous integration (Jenkins) into the deployment lifecycle.
- Migrated critical on-premise databases to managed cloud SQL/NoSQL solutions, improving read throughput by 3x.

TECHNICAL SKILLS

System Architecture, Distributed Systems, Cloud-Native Design, Java / Spring Boot,
AWS (EC2, EKS, S3, RDS), Docker & Kubernetes, Kafka & Event-Driven, NoSQL & PostgreSQL,
DevOps & CI/CD, Agile Leadership

EDUCATION

West Bengal University of Technology

2015

Bachelor of Technology (B-Tech), Computer Science and Engineering